

March 28, 2025

Andrew S. Johnston
Executive Secretary
Maryland Public Service Commission
William Donald Schaefer Tower
6 St. Paul Street, 16th Floor
Baltimore, Maryland 21202

Re: Case No. 9478; Comments of VGIC on Utility Managed Charging Programs

Dear Mr. Johnston,

The Vehicle-Grid Integration Council (“VGIC”)¹ respectfully provides its comments in response to Potomac Electric Power Company’s (“PHI”) and Delmarva Power & Light Company (“Delmarva Power”; collectively with PHI the “PHI Utilities” or “Utilities”) Load Management Programs in their Phase II Electric Vehicle Portfolio to the Maryland Public Service Commission (“Commission”) in Case No. 9478. VGIC also respectfully provides comments on Baltimore Gas and Electric (“BGE”) and the Potomac Edison Company (“Potomac Edison”) respective EV Phase II Proposals in Case No. 9478.

I. Introduction

VGIC is a 501(c)(6) membership-based advocacy group committed to advancing the role of electric vehicles (“EVs”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring the value from EV deployments and flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid.

II. VGI Programs as a Part of the Phase II EV Portfolios

VGIC appreciates the PHI Utilities, BGE, and Potomac Edison’s work on advancing managed charging. These utilities have been leaders in making the transition to electric vehicles affordable, reliable, and clean.

The PHI Utilities are proposing two load management programs, an EV-TOU rate and Smart Charge Management (“SCM”; referred to collectively with EV-TOU as the “EV Load Management Programs”). The EV Load Management Programs aim to reduce the cost of service

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., Fermata Energy, Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, BorgWarner, ChargeScape, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, Gravity, GridBeyond, GridWiz, Hyundai Cradle, Innovation Core SEI, Kaluza, Landis + Gyr, LG Innotek, Mercedes-Benz, Orange Charger, Peak Power, PowerFlex, Stellantis N.V., Sunrun, Tesla, The Mobility House, Toyota Motor North America, Inc., WeaveGrid, Clean Power Alliance, Holy Cross Energy, Los Angeles Department of Water & Power, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>)

related to EV load, create net benefits for all of the PHI Utilities' customers, and further encourage the electrification of transportation through lowering the cost of charging.

BGE is proposing a Charge Anywhere program to incentivize off-peak charging for EV drivers without dedicated charging access. Shifting EV charging to off-peak periods reduces the cost of service, enhances reliability, and creates financial savings for EV drivers who live in multi-family dwellings. BGE is also proposing an EV Energy Management System (“EMS”) pilot to help residential customers avoid the cost of upgrading their electric service panels when installing a level 2 charger, thereby reducing the financial barriers to electrification.

Potomac Edison is proposing a Managed Charging Pilot that optimizes the charging schedule of participants to reduce their cost of service.

The PHI Utilities, BGE, and Potomac Edison are all helping to realize Maryland’s commitment to clean energy and the electrification of transportation through the EV Load Management Programs. The programs will aid in Maryland’s achievement of the Advanced Clean Cars II rule (“ACC II”), which requires 100% of passenger car and light truck sales to be EVs by vehicle model year 2035. The Programs also further the policy goals of the Maryland Climate Solutions Now Act of 2022, which targets a 60% reduction in greenhouse gas emissions by 2031, along with a goal to reach net-zero by 2045.

III. PHI Utilities - Cost Effectiveness

In their Phase II application, the PHI Utilities filed a cost-effectiveness analysis conducted by Brattle (“Brattle analysis”). VGIC believes the full context of the analysis should be taken into consideration and that the net benefits of managed charging are significantly higher than it may seem.

The EV-TOU and SCM programs are essentially identical to programs operated by Baltimore Gas and Electric (BGE). BGE filed a combined cost-effectiveness of these programs that showed a score of 1.58.²

One possible explanation of the difference between BGE and the PHI Utilities is that Brattle did not use actual load shapes from the SCM program. Instead, the Brattle analysis relied on estimated load shifts from a passive pilot program in San Diego Gas & Electric (“SDG&E”)³. As a result, the analysis may not reflect charging behavior in Maryland, where drivers likely have different patterns and behaviors for both TOU and active managed charging programs. We believe the estimated benefits would be higher using Maryland data.

VGIC also believes the distribution benefits of SCM could be higher than included in the Brattle analysis. Many avoided cost approaches take a simplified approach to calculating the distribution benefits of a program, by basing the benefits on the reduction in the peak of the bulk electric system. However, SCM aims to reduce peaks in the distribution system that may occur outside of when the bulk system is capacity constrained.

² see *BGE Smart Charge Management Program Proposal*, filed in Case No. 9478 on June 4, 2024, p. 9

³ see footnote 62 in the Brattle Report

According to the Smart Electric Power Alliance (“SEPA”), “Distribution effects of EV charging often manifest at lower levels of EV adoption than those at the bulk level. These impacts are closely related to the number of customers sharing a residential transformer and the size of that transformer. When transformers serve between 5-10 customers and are only rated between 10-34 kVA, coincident EV charging can quickly overload circuits, not even taking into account other loads on the circuit.” In other words, EV charging can overload many distribution assets during hours that are considered “non-peak” for the purposes of the generation and transmission system.

SCM aims to protect distribution assets, such as secondary transformers, from localized peaks primarily caused by EV charging which can often occur during “non-peak” hours⁴. However, the distribution benefits of SCM are not properly valued using the traditional approach of only looking at peak reduction at the bulk level.

Based on independent reports from other states, VGIC expects there to be significant additional benefits of SCM compared to EV-TOU alone. For example, a US Department of Energy study entitled “Multi-State Transportation Electrification Impact Study” released in March 2024 explored potential infrastructure costs related to EV load in five states - California, Illinois, New York, Oklahoma, and Pennsylvania. The study found that “the incremental distribution grid capital investment of \$2.3 billion estimated by this study is reduced 30% to \$1.6 billion when [Plug-in electric vehicles or “PEVs”] charging loads at home and depot locations are managed [...] within the context of this study, the [managed charging] strategy is agnostic to non-PEV residential, commercial, and industrial loads on the distribution network (meaning simulated PEV loads are not optimized relative to non-PEV loads). Accordingly, the results present a conservative estimate of the potential distribution grid savings from managing charging load locally.”⁵

PHI’s load management programs have been effective in shifting load away from peak periods, as well as reducing the coincident charging of EVs. Given the promising cost-effectiveness results of EV-TOU and SCM conducted by BGE, as well as evidence from other states, VGIC urges the Commission to continue expanding the deployment of the load management programs.

V. BGE - Charge Anywhere Program

BGE’s Charge Anywhere program will provide financial incentives to customers living in multi-family dwellings to charge during off-peak times. This program will expand the scope of customers who can participate in managed charging programs, especially given that roughly 25% of BGE’s customers live in multi-family dwellings. The Charge Anywhere program supports

⁴ There are two definitions of “off-peak” that could be used in this case. The hour can be off-peak based on a TOU schedule or off-peak in terms of these hours not having constrained capacity for the generation and transmission system. Both definitions would work in this context, since local timer peaks can occur either during TOU off-peak hours or when there is plenty of generation and transmission capacity.

⁵ <https://www.energy.gov/sites/default/files/2024-03/2024.03.18%20NREL%20BNL%20Kevala%20DOE%20Multi-State%20Transportation%20Electrification%20Impact%20Study%20FINAL%20DOCKET.pdf>

electrification, providing participants with a stronger financial incentive to utilize an EV, while also reducing their cost of service. VGIC strongly supports the Charge Anywhere program.

VI. BGE – EV EMS Demonstration

EV customers often seek a level 2 charger to be installed in their home. However, installing such a charger may require service upgrades which can cost up to \$6,000. This high upfront cost can be prohibitive for many households to consider purchasing an EV.

BGE's EMS demonstration will manage load to avoid the need for service upgrades, thus significantly reducing the upfront cost of electrification. Participants would also be enrolled in SCM and/or EV TOU to further optimize charging schedules and reduce the cost of charging. Therefore, the EV EMS demonstration aims to create cost savings for EV adoption, EV ownership, and the entire electric system. VGIC strongly supports BGE's EV EMS Demonstration, as this type of flexible connection is a critical element of a well-balanced vehicle-grid integration portfolio.

VI. Potomac Edison - Managed Charging Pilot

Potomac Edison's proposed managed charging pilot has similar goals and design to BGE and the PHI Utilities SCM program. As those programs have been a successful and cost-effective way to capture a wide range of benefits - enhancing the affordability of EV charging, reducing utility costs, and increasing the reliability of the distribution system - VGIC strongly supports Potomac Edison's managed charging pilot.

VII. Conclusion

The Phase II proposals submitted by the PHI Utilities, BGE, and Potomac Edison all represent significant advancements in the electrification of transportation in a way that creates benefits for all Maryland ratepayers and residents. We recommend the Commission approve these pilots and the expansion of these programs to ensure Maryland continues to lead on transportation electrification, while protecting grid reliability and delivering customer value.

VGIC appreciates the opportunity to provide comments and looks forward to further collaboration with stakeholders on these important initiatives.